

Identifying Organizational Identification as a Basis for Attitudes and Behaviors: A Meta-Analytic Review

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Organizational identification has been argued to have a unique value in explaining individual attitudes and behaviors in organizations, as it involves the essential definition of entities (i.e., individual and organizational identities). This review seeks meta-analytic evidence of the argument by examining how this identity-relevant construct functions in the nexus of attitudinal/behavioral constructs. The findings show that, first, organizational identification is significantly associated with key attitudes (job involvement, job satisfaction, and affective organizational commitment) and behaviors (in-role performance and extra-role performance) in organizations. Second, in the classic psychological model of attitude-behavior relations (Fishbein & Ajzen, 1975), organizational identification is positioned as a basis from which general sets of those attitudes and behaviors are engendered; organizational identification has a direct effect on general behavior above and beyond the effect of general attitude. Third, the effects of organizational identification are moderated by national culture, a higher-level social context wherein the organization is embedded, such that the effects are stronger in a collectivistic culture than in an individualistic culture. Theoretical and practical implications of the findings and future research directions are discussed.

Keywords: organizational identification, attitude-behavior relations, culture, meta-analysis

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Every entity needs to have a fundamental sense of identity—“Who am I?” or “Who are we?”—to appropriately and meaningfully relate to others and surrounding social contexts (Gioia, 1998; Jenkins, 2008). In a salient social domain in modern society (i.e., organizations; Haslam, 2004; Hogg & Terry, 2000), organizational identification reflects this sense of identity by capturing a psychological state wherein one defines one’s self by the same attributes that one believes define one’s organization (Dutton, Dukerich, & Harquail, 1994; Pratt, 1998). Organizational identification has often been highlighted as central because it involves the very

essential definition of entities (i.e., individual and organizational identities), constituting a fundamental subtext based on which attitudes and behaviors in organizations are developed (Albert, Ashforth, & Dutton, 2000; Hogg & Terry, 2001; van Knippenberg & van Schie, 2000). Recognizing its importance, a large amount of research has revealed numerous correlates of organizational identification, and several qualitative reviews (Ashforth, Harrison, & Corley, 2008; Haslam & Ellemers, 2005; Pratt, 1998; van Dick, 2004) and a meta-analysis (Riketta, 2005) summarized those correlates. However, merely accumulating correlates does not necessarily offer evidence of the unique value of this identity-relevant construct (Ashforth et al., 2008). This study thus aims to provide systematic evidence of the implications of organizational identification by positioning it in the framework of attitude-behavior relations (Ajzen & Fishbein, 1977; Bentler & Speckart, 1981; Fishbein & Ajzen, 1975; Glasman & Albarracín, 2006), a classic theoretical lens in the psychology literature. By comparing alternative structural models, we examine how organizational identification functions in relation to work attitudes (job involvement, job satisfaction, and affective organizational commitment) and behaviors (in-role performance and extra-role performance), finally providing meta-analytic evidence that organizational identification uniquely affects behaviors in organizations, above and beyond the effects of attitudes.

In doing so, we further extend the literature by examining whether and how the effects of organizational identification on attitudes/behaviors vary across different national cultural contexts.

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Researchers have noted that the cultural value system espoused in a certain national context has critical implications for the effects of identity/identification in the organization (Baker, Carson, & Carson, 2009; Erez & Earley, 1993), but no cumulative evidence, to our knowledge, exists in the literature. Building upon the notion from cross-cultural research that national culture provides considerable inputs in shaping individual attitudes and behaviors (Gelfand, Erez, & Aycan, 2007; Taras, Kirkman, & Steel, 2010), we expect that the effects of organizational identification will vary depending on whether the national culture, wherein the organizational setting is embedded, concurrently values pursuing an identity overlap between an individual and a social collective (i.e., organization). In this study, we thus seek for meta-analytic evidence of how national culture moderates the relations between organizational identification and its attitudinal/behavioral outcomes. We believe that our consideration of national culture will substantially contribute to the organization identification literature by providing a comprehensive picture of organizational identification effects.

The Effects of Organizational Identification

As a dominant psychological approach to identity and identification, social identity theory explains how individuals construct their self-concepts from the identity of the collectives they belong to (Tajfel, 1978; Tajfel & Turner, 1985). Tajfel (1978) defined social identity as “that part of an individual’s self-concept which derives from his knowledge of his membership of a social group (or groups) together with the value and emotional significance attached to that membership” (p. 63). Social identities are shared by members and accentuate members’ perceived similarity. Members share the group’s prototypical traits, and thus depersonalize their self-concepts (Hogg, Terry, & White, 1995; Turner, Hogg, Oakes, Reicher, & Wetherell, 1987); through this process of categorizing the self into a more inclusive social entity, “*I becomes we*” (Brewer, 1991, p. 476).

As a salient social domain in modern society, organizations provide a significant social identity (Haslam, 2004; Hogg & Terry, 2000). Organizational identification thus roots individuals in the organization, leading organizational attributes such as espoused values, goals, and norms to become salient and self-defining for individuals; through organizational identification, the identity boundary between individual and organization becomes blurred. In turn, an organizationally identified employee, as a “microcosm of the organization” (Ashforth et al., 2008, p. 333), is likely to have attitudes and take actions that benefit the whole organization rather than benefitting individual self-interest (Ashforth & Mael, 1989; Haslam & Ellemers, 2005; Pratt, 1998; van Knippenberg, 2000). To illustrate, when a Google employee describes herself as creative and innovative which are the attributes she ascribes to the Google organization (i.e., categorizing herself as a “Googler”), it reflects her organizational identification, and she is likely to think, feel, and behave in ways that are expected among prototypical Googlers. Below, we detail the theoretical rationales for how an individual’s work attitudes and behaviors are shaped by organizational identification.

Attitudinal Outcomes

Researchers have noted that organizational identification has significant impacts on individual attitudes in organizations. We focus on job involvement, job satisfaction, and affective organizational commitment, the three classic work attitudes in the literature (Brooke, Russell, & Price, 1988; Newman & Harrison, 2008; Newman, Joseph, & Hulin, 2010).

Job involvement and job satisfaction. Job involvement refers to the “degree to which an employee psychologically relates to his or her job and to the work performed therein” (Cooper-Hakim & Viswesvaran, 2005, p. 244), and job satisfaction refers to an “evaluative state that expresses contentment with, and positive feelings about, one’s job” (Judge & Kammeyer-Mueller, 2012, p. 347). These two job attitude constructs are distinguishable in that job involvement reflects a cognitive belief state related to one’s job, while job satisfaction reflects an emotional state of liking the job that one performs (Brooke et al., 1988; Mathieu & Farr, 1991).

Job involvement literature suggests that an individual tends to be more cognitively connected to one’s job when the job is perceived to be fulfilling one’s psychological needs (Kanungo, 1979, 1982). In this vein, researchers have proposed that individuals become more involved in their jobs when the jobs are designed to provide a sense of the meaningfulness—the sense that one’s contribution significantly influences the overall effectiveness of the organization (Brown, 1996; Hackman & Lawler, 1971; Hackman & Oldham, 1980). Along the same line of reasoning, individuals with high organizational identification are more likely to find their jobs meaningful because, with identity merging between individual and organization (van Knippenberg & Sleebos, 2006) or depersonalization (Brewer, 1991; Hogg et al., 1995), they are likely to sense that fulfilling their individual jobs simultaneously means contributing to the achievement of the organization’s objectives and missions. Accordingly, strong identifiers are more likely to be psychologically connected to or involved in their jobs, and several previous studies indeed have shown that organizational identification is positively associated with job involvement (e.g., Hassan, 2010; van Knippenberg & van Schie, 2000). We thus expect that organizational identification will increase job involvement.

Organizational identification is also likely to enhance an individual’s emotional response to one’s job. Individuals have a basic need to view themselves positively (Allport, 1955; Steele, 1988) and actively pursue information to enhance their self-concepts (Baumeister & Jones, 1978; Tesser, 1988). For individuals with high organizational identification, the identities of the individual and the organization overlap, and thus they are inclined to seek positive aspects of their organization to attain a positive self-concept (Bergami & Bagozzi, 2000; Dutton et al., 1994; Pratt, 1998). Their positive evaluation of the organization is, in turn, likely to lead them to feel satisfied with their jobs because job satisfaction is developed from one’s interpretation of the job circumstances (Brief, 1998) and strong identifiers tend to feel that they perform their jobs within the favorable job circumstance—the organization (van Dick et al., 2004a). In accordance with this logic, several previous studies have found that identification with the organization is positively related to the feeling of satisfaction with one’s job (e.g., van Dick et al., 2004a; van Dick, van Knippenberg, Kerschreiter, Hertel, & Wieseke, 2008; van Knippenberg

& van Schie, 2000). Hence, we expect that organizational identification will increase job satisfaction.

Affective organizational commitment. Affective organizational commitment refers to an individual's psychological bond with the organization constructed through an affective attachment to the organization and internalization of the organization's values and goals (Judge & Kammeyer-Mueller, 2012). Affective organizational commitment bears a similarity to organizational identification because both regard an individual's psychological attachment to the organization (O'Reilly & Chatman, 1986; Riketta & van Dick, 2005). However, commitment refers to a more general attitude constructed based on the perception of social exchange between individual and organization, two *separate* psychological entities; thus, it does not contain a self-defining nature, which is central in the concept of identification, thereby not reflecting psychological *oneness* (Ashforth & Mael, 1989) between individual and organizational identities (Ashforth et al., 2008; Cole & Bruch, 2006; Klein, Molloy, & Brinsfield, 2012; Pratt, 1998; van Knippenberg & Sleebos, 2006). Accordingly, researchers have suggested that it is reasonable to examine affective organizational commitment as an attitudinal outcome of organizational identification; as individuals develop a sense of oneness or identification with the organization, they build emotional bonds with and engagement in the organization, which leads them to be more committed to it (Bergami & Bagozzi, 2000; Meyer, Becker, & Vandenberghe, 2004; Meyer, Becker, & van Dick, 2006). In other words, identification is "the appropriation of identity" and commitment is "the binding to action" (Cheney & Tompkins, 1987, p. 9), which implies that possessing a social identity (i.e., identification) is a basis or a precursor for developing commitment to the corresponding collective (Meyer et al., 2006; Meyer & Herscovitch, 2001). In accordance with the empirical evidence supporting this theoretical view (e.g., Bergami & Bagozzi, 2000; Marique & Stinglhamber, 2011), we expect that organizational identification will increase affective organizational commitment.

Behavioral Outcomes

Previous research suggests that organizational identification also influences behaviors as well as attitudes in organizations.

In-role performance. Individuals with high organizational identification are more likely to contribute to collective goals (i.e., organizational goals) by taking actions that benefit the whole organization (Haslam & Ellemers, 2005). One way of reifying behavioral responses to organizational identification is to successfully enact in-role behaviors that are required by formal job descriptions, directly serving the goals of the organization (Foote, 1951; van Knippenberg, 2000). Put differently, strong identifiers are likely to exert substantial individual efforts for the sake of organizational goals, and those efforts translate into their intrinsic motivation to perform well in their individual jobs, resulting in enhanced job performance (Haslam & Ellemers, 2005; van Knippenberg, 2000; van Knippenberg & van Schie, 2000). For instance, studies show that strong identifiers tend to share information and communicate with coworkers (Grice, Gallois, Jones, Paulsen, & Callan, 2006) and make choices that serve the organization's strategic interests (Bartel, 2001; Tompkins & Cheney, 1985). Also, strong identifiers are more likely to engage in customer-oriented activities that critically contribute to organizational effectiveness

(Farrell & Oczkowski, 2012; Homburg, Wieseke, & Hoyer, 2009). Based on all of this reasoning and evidence, we expect that organizational identification will increase in-role performance.

Extra-role performance. Another notable behavioral outcome of organizational identification is extra-role performance—discretionary actions that are beyond formal job descriptions and not directly recognized by an organization's formal reward system (Konovsky & Pugh, 1994; Podsakoff, MacKenzie, Paine, & Bachrach, 2000) such as employee voice behaviors (Van Dyne & LePine, 1998) and organizational citizenship behaviors (OCBs; Organ, 1997). For strong identifiers, voluntarily helping the organization achieve its goal (e.g., adhering to informal company rules, attending company meetings that are not mandatory) is important because the organization's goal is theirs as well. In addition, for strong identifiers, other organizational members who comprise a significant part of their perceptions of what the organization is provide substantial meanings for their self-definitions (Sluss & Ashforth, 2007, 2008). Strong identifiers thus are likely to perceive that helping other organizational members through extra-role behaviors (e.g., willingly assisting others experiencing work-related problems, spending time helping newcomers) parallels helping themselves (Dukerich, Golden, & Shortell, 2002; van Dick, Grojean, Christ, & Wieseke, 2006). In sum, through organizational identification, they become good organizational citizens (Bergami & Bagozzi, 2000; van Dick et al., 2006). Previous studies have offered strong empirical support for this reasoning (Christ, van Dick, Wagner, & Stellmacher, 2003; Dukerich et al., 2002; Johnson, Morgeson, & Hekman, 2012; van Dick et al., 2006), and therefore we expect that organizational identification will increase extra-role performance.

Uniqueness of organizational identification's effects on behaviors. Those theoretical and empirical notions of the link between organizational identification and in-role/extra-role performance indicate that organizational identification influences behaviors not necessarily through attitudes. Some researchers have specifically argued that, different from attitudes which primarily capture relatively instrumental, ephemeral, or short-term motivators of behavior, identity and identification constitute relatively long-term or fundamental motivators (Chang, Piliavin, & Callero, 1988; Haslam & Ellemers, 2005; Sparks, 2000). In organizational settings, thus, when individuals identify with the organization, their identity label (Biddle, Bank, & Slavings, 1987) or identity standard (Stets & Burke, 2000) of a prototypical organizational member, as "agent or cause of behavior" (Burke & Reitzes, 1981, p. 83), is likely to lead them to exhibit behaviors that are favored in the specific organizational context (Terry & Hogg, 1996; Terry, Hogg, & White, 1999). In accordance with this notion, several empirical studies (e.g., Biddle et al., 1987; Chang et al., 1988) have shown that the effects of identity/identification on behaviors persist even after the effects of attitudes have been accounted for. However, other researchers have indicated that the effects of organizational identification on behaviors are not uniquely differentiated from those of attitudes because organizational identification conceptually overlaps with other attitude constructs—*affective organizational commitment* in particular (O'Reilly & Chatman, 1986; Riketta & van Dick, 2005). This alternative perspective comes from the notion that work attitudes are generally defined as cognitive and emotional evaluations of work experiences (Harrison, Newman, & Roth, 2006; Judge & Kammeyer-

Mueller, 2012) and organizational identification also involves evaluation of work experiences—an individual's cognitive belief in and emotional attachment to the organizational membership (Ashforth et al., 2008; Edwards, 2005; Harquail, 1998).

We test the validity of those two perspectives by formulating two alternative meta-analytic path models. According to the compatibility principle (Ajzen, 1988; Ajzen & Fishbein, 1977), attitudes lead to behaviors only when the measures of attitudes and behaviors are matched in their action, target, context, and time. Invoking this principle, researchers (Harrison et al., 2006; Newman et al., 2010) have shown that a *general set of attitudes* (i.e., a higher-order aggregate of work attitudes which connotes a broad evaluation of work experiences) provides maximal explanatory power when predicting a *general set of behaviors* (i.e., a higher-order aggregate of work behaviors broadly deemed effective in organizations) compared with when predicting each specific behavior separately. Accordingly, if there is a considerable conceptual overlap between organizational identification and other attitude constructs, a latent variable of general attitude encompassing job involvement, job satisfaction, affective organizational commitment, and organizational identification will be maximally predictive of a latent variable of general behavior which includes in-role performance and extra-role performance (Model 1 in Figure 1). However, if organizational identification is a unique construct that is distinctive from other attitude constructs, the other path model (Model 2 in Figure 2) that specifies organizational identification as a separate variable which functions as a basis influencing general behavior not only indirectly—through general attitude—but also directly—above and beyond the effect of general attitude—will have higher explanatory power than Model 1.

National Culture as a Moderator of Organizational Identification Effects

For a more comprehensive understanding of the attitudinal/behavioral effects of organizational identification, we further examine the moderator of these effects: national culture. National culture shapes core values and norms shared and transmitted within the specific national context, thus providing the “collective programming of the mind” (Hofstede, 1984, p. 389) that controls individuals' responses in the given context. According to culture–fit theory (Aycan, Kanungo, & Sinha, 1999; Kanungo & Jaeger, 1990), national culture, as a social context at the higher level,

manifests its cultural value system in, and hence constraining, organizational policies and practices (Erez & Earley, 1993; Hofstede, 1983; Johns, 2006; Rabl, Jayasinghe, Gerhart, & Kühmann, 2014), which in turn influence the attitudes and behaviors of individuals embedded in these layers of contexts (Gelfand et al., 2007; Taras et al., 2010). In this sense, by noting that “the validity of [management] theories may stop at national borders” (p. 82), Hofstede (1993) argued that national culture is a significant boundary condition of the phenomena around organizations and the individuals within them. Building on those previous notions, we expect that the effects of organizational identification on attitudes/behaviors will be moderated by national culture; when pursuing the individual–organization identity connection is aligned with the cultural value system espoused in a specific national context, the organizational identification effects will be strengthened.

Hofstede (1980, 2001) proposed five value dimensions that characterize cultural variability: individualism/collectivism, time-orientation, uncertainty-avoidance, power-distance, and masculinity/femininity. Among those five, previous theoretical and empirical considerations have suggested that the first three dimensions may have particular implications regarding organizational identification (Baker et al., 2009; Clugston, Howell, & Dorfman, 2000). Therefore, we take these three dimensions—individualism/collectivism, time-orientation, and uncertainty-avoidance—into consideration.

Regarding individualism/collectivism, research shows that cultures differ in the extent to which they highlight an individual's connectedness to collectives; individualistic culture prioritizes individual goals over the goals of collectives (e.g., an organization), whereas a collectivistic cultural value system makes less distinction between individual and collective goals (Hofstede, 1980, 2001; Triandis, 1995). Accordingly, in an individualistic culture, self-concept is defined in an independent way with an emphasis on an individual's unique traits, and thus each individual is likely to value their independence in thinking, feeling, and acting as a unique entity. In contrast, in a collectivistic culture, self-concept is defined in an interdependent way with substantial influences of others in one's social relationships, and thus individuals are likely to value attitudes and behaviors organized with reference to the social collectives to which they belong (Erez & Earley, 1993; Markus & Kitayama, 1991). In this vein, defining one's self in terms of the organization, a significant social collective in modern

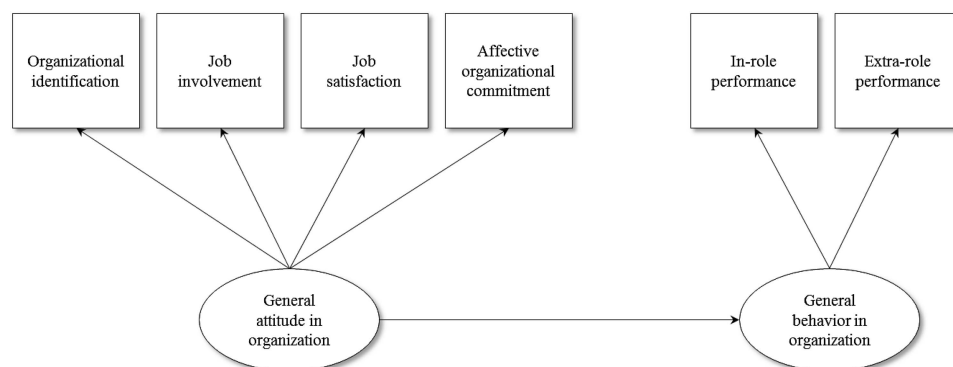


Figure 1. Model 1: Organizational identification as a component of general attitude.

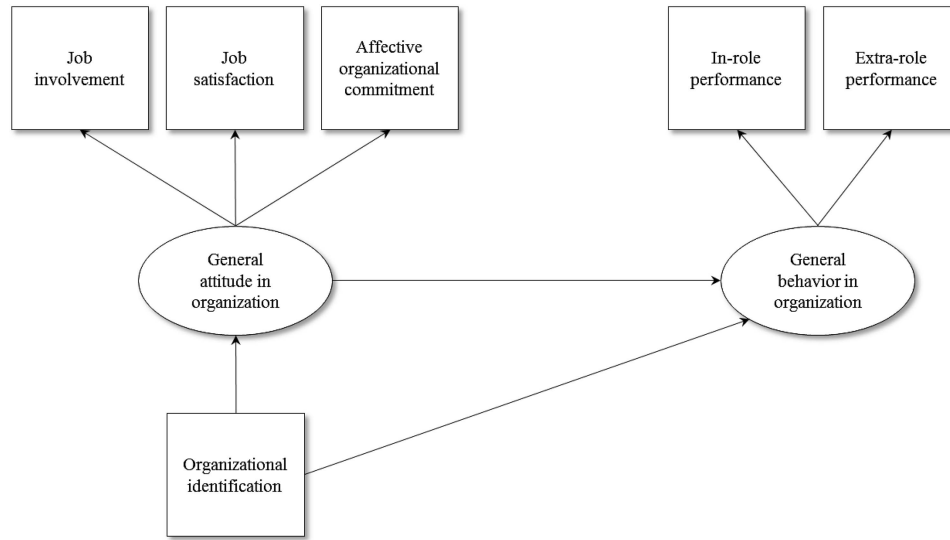


Figure 2. Model 2: Organizational identification as a basis for general attitude and behavior.

society, is more congruent with the collectivistic than with the individualistic value system. Accordingly, we expect that organizational identification will have more salient effects on the work attitudes/behaviors of individuals in a collectivistic than in an individualistic culture. Supporting this, empirical evidence shows that psychological connection to the organization is more influential on work behaviors such as OCBs for those in a collectivistic culture than those in an individualistic culture (Felfe, Yan, & Six, 2008; Wasti, 2003).

The other two dimensions, time-orientation and uncertainty-avoidance, are a less salient focus of empirical investigation, but they have implications for the effects of organizational identification. Time-orientation of a culture reflects a preference for past, present, or future thinking. In a short-term-oriented culture, individuals tend to pursue quick results and have weak propensities to save for the future. In a long-term-oriented culture, however, individuals exhibit a strong propensity to save and endure in attaining results (Hofstede, 2001; Hofstede & Bond, 1988). In terms of employment relationships, individuals in a short-term-oriented cultural context are likely to seek immediate benefits by moving across organizations, whereas those in a long-term-oriented cultural context may value long-term relationships with employing organizations, focusing on future achievements (Baker et al., 2009; Schalk & Soeters, 2008). Thus, we expect that in a long-term-oriented culture, having a deep psychological connection to an organization through organizational identification, in alignment with the cultural value system that highly regards long-term employment relationships, will have greater effects on work attitudes and behaviors than in a short-term-oriented culture.

Uncertainty-avoidance reflects a society's endurance for ambiguity and uncertainty. In an uncertainty-avoidance cultural context, individuals feel threatened by ambiguous situations and unpredictable environments (Hofstede, 1980), and thus are less inclined to leave their organizations to pursue job security and stabilized rules/values (Chew & Putti, 1995; Clugston et al., 2000). As a means of maintaining their present environments and posi-

tions, individuals in this context are likely to feel a sense of obligation to and identification with their current employer and pursue long tenure to minimize potential anxiety and loss (Baker & Carson, 2011; Schalk & Soeters, 2008). Based on this logic, we expect that, compared with a low uncertainty-avoidance culture, the organizational identification effects will be more salient in a high uncertainty-avoidance culture where the cultural value system supports the rationale for defining one's self in terms of the stable social entity—the organization.

Method

Literature Search

We employed multiple search strategies to identify all relevant English-written articles published before or during April, 2014. First, we searched the EBSCO, JSTOR, PROQUEST, ISI Web of Knowledge, PsycINFO, and Google Scholar databases using the keyword *identification* or *identity* in combination with *organization* or *company* and other keywords such as *membership* and *categorization*. Second, we looked through the reference sections of all qualitative and quantitative reviews on organizational identification (e.g., Ashforth et al., 2008; Edwards, 2005; Riketta, 2005). Third, using the cited reference search offered by the ISI Web of Knowledge, we searched among publications that had cited important articles in the field (e.g., Ashforth et al., 2008; Ashforth & Mael, 1989; Dutton et al., 1994; Mael & Ashforth, 1992; Pratt, 1998; Riketta, 2005). Fourth, we searched online for journals with organizational identification studies still in press (e.g., *Psychological Bulletin*, *Journal of Applied Psychology*, *Academy of Management Journal*, *Organization Science*, *Administrative Science Quarterly*, *Personnel Psychology*, *Journal of Management*, *Journal of Organizational Behavior*, *Organizational Behavior and Human Decision Processes*, *International Journal of Human Resource Management*). Finally, in an attempt to find unpublished manuscripts, we contacted authors who had recently

published in the area of organizational identification, and we also announced a call for unpublished manuscripts on organizational researchers' e-mail list service such as the HR (human resources) Listserv and the OB (organizational behavior) Listserv. Those combined efforts yielded 341 published articles and 16 unpublished manuscripts and dissertations.

Inclusion Criteria and Sample

To be included in the meta-analysis, a study had to satisfy the following four criteria. First, the study had to address an *organizational* identification issue, not other types of identification issues, such as occupational identification, professional identification, and subgroup/team identification (e.g., Täuber & Sassenberg, 2012; Wann, Waddill, Polk, & Weaver, 2011). Second, the study had to be an empirical one that reported correlations between organizational identification and individual-level outcome variables. We thus excluded theory or review papers and empirical papers that used qualitative methods because they lacked the necessary information on the correlations between organizational identification and outcomes. Ninety-one articles were excluded for those two reasons. Third, the study had to report the correlation(s) between organizational identification and its attitudinal and/or behavioral outcome(s). Therefore, studies were excluded when they only include correlations between organizational identification and its antecedents (e.g., organizational prestige, perceived organizational support) or correlations between organizational identification and its outcomes other than attitudinal or behavioral ones (e.g., emotion, burnout, turnover intention). In this process, 99 articles were excluded. Fourth, the study had to measure organizational identification using the scales that emphasize an individual's oneness perception or the self-defining nature of identification. A representative example is the Mael scale (Mael & Ashforth, 1992). This scale has long been known to measure the oneness perception of organizational identification, partly because of its association with the seminal article by Ashforth and Mael (1989), and thus has been the most widely used in the organizational identification literature (Haslam, 2004). The Mael scale includes items such as "When someone criticizes [name of organization], it feels like a personal insult," and "When I talk about [name of organization], I usually say 'we' rather than 'they.'" The scale developed by van Dick, Wagner, Stellmacher, and Christ (2004b) has also been widely used, as it contains the self-defining nature of identification. This scale includes items such as "I identify myself as a member of [name of organization]," and "Being a member of [name of organization] reflects my personality well." Another exemplifying scale is the graphical Venn diagram scale which measures organizational identification as the degree of overlap between two circles that denote a respondent's own identity and his or her organization's identity (e.g., Bergami & Bagozzi, 2000). The studies included in our analysis, in which organizational identification was measured using the scales other than the Mael scale, are listed in Table 1. In some organizational identification scales (e.g., the Organizational Identification Questionnaire; Cheney, 1983), however, the operationalization of identification is similar to that of commitment, and thus they are not clearly distinguishable from organizational commitment scales such as the Organizational Commitment Questionnaire (Mowday, Steers, & Porter, 1979) and the Affective Commitment Scale

(Allen & Meyer, 1990). Accordingly, we excluded studies using the identification scales similar to commitment scales, thus not adequately measuring the self-defining nature of identification. In this process, 18 articles were excluded from our meta-analysis. Table 1 also presents the list of those excluded studies.

After applying all the inclusion criteria, we obtained an initial data set of 149 organizational identification–outcome correlations from 114 studies in 86 articles. The Appendix provides a summary of the studies and samples used in the meta-analysis, and the complete list of articles that were considered but ultimately excluded is in the online supplemental material.

To calculate the overall correlation, we coded all possible zero-order correlations from each study. Of the coded correlations, some of the correlations were nonindependent because they were computed from the same sample. Thus, when correlations were based on multiple measures of the same criterion in the same sample, such as intrinsic satisfaction and extrinsic satisfaction in Becker's (1992) study, we combined the multiple measures into a composite using the composite formulas (Ghiselli, Campbell, & Zedeck, 1981, pp. 163–164). Fifty nonindependent correlations were combined for this reason.

Yet, many studies still yielded more than one relevant correlation because we were interested in the relations between organizational identification and several different types of individual outcomes, such as job satisfaction, affective organizational commitment, and in-role performance. In such cases, adopting the approach used in the previous meta-analysis studies (e.g., Albaracín et al., 2005; Albaracín, Johnson, Fishbein, & Muellerleile, 2001; Baas, De Dreu, & Nijstad, 2008; Durantini, Albaracín, Mitchell, Earl, & Gillette, 2006), we allowed more than one correlation per study to be included in our final sample; that is, even after combining correlations for reasons of independence, we still used multiple correlations from one study if they concerned different types of outcomes—job satisfaction and in-role performance in Ashforth, Sluss, and Saks' (2007) study, for example. Although multiple correlations were used from one study, we ensured that, in the analysis of the correlation between organizational identification and each specific outcome, we use only one correlation from one study, thus not violating the sample independence assumption (Cooper & Hedges, 1994). However, in the meta-analysis of the overall correlation between organizational identification and all outcomes, this approach could still violate the independence assumption. Hence, when we analyze this overall correlation, we used one correlation (i.e., the composite correlation) per study. Table 2 shows the stem-and-leaf display of the 114 independent correlations, and Figure 3 shows the funnel plot of the correlations. The funnel plot provides initial evidence that selection or publication bias is unlikely to be present because the distribution of sample is symmetrical and the form of a funnel centers on the mean effect size, with smaller variability as the sample size increases. We present more formal publication bias tests below.

Coding of Studies

The studies were coded independently by two coders for correlation, sample size, reliability estimates, type of organizational identification outcome, national culture, and study characteristics (e.g., data structure, publication status). The initial intercoder

Table 1
Inclusion Decisions for the Studies Not Using the Mael Scale

Organizational identification scale type	Study	Measure source	Inclusion decision
Newly developed scales since the Mael scale	Gümüş et al. (2012)	van Dick et al. (2004b)	Included
	Michel et al. (2010)	van Dick et al. (2004b)	Included
	van Dick et al. (2004a)	van Dick et al. (2004b)	Included
	van Dick et al. (2006) Samples 1 to 7	van Dick et al. (2004b)	Included
	van Dick et al. (2007)	van Dick et al. (2004b)	Included
	van Dick et al. (2008)	van Dick et al. (2004b)	Included
	Smidts et al. (2001)	Smidts et al. (2001)	Included
	Walumbwa et al. (2009)	Smidts et al. (2001)	Included
	Walumbwa et al. (2011)	Smidts et al. (2001)	Included
	Zhao et al. (2014)	Smidts et al. (2001)	Included
	Edwards & Peccei (2010)	Edwards & Peccei (2007)	Included
	Fuchs & Edwards (2012)	Edwards & Peccei (2007)	Included
	Peters et al. (2010)	Doosje et al. (1995)	Included
	Richter et al. (2006)	Doosje et al. (1995)	Included
	Norman et al. (2010)	Avey et al. (2008)	Included
	Christ et al. (2003)	Christ et al. (2003)	Included
	Johnson et al. (2012)	Johnson et al. (2012)	Included
	Stoner & Gallagher (2011)	Stoner, Perrewé, & Hofacker (2011)	Included
	Amiot et al. (2006)	Terry & Hogg (1996)	Included
	Hassan (2010)	Tyler & Blader (2000)	Included
Graphical scales	Bartel (2001)	Bergami & Bagozzi (2000)	Included
	Korschun et al. (2011)	Bergami & Bagozzi (2000)	Included
	Mayfield & Taber (2010)	Shamir & Kark (2004)	Included
	Wolfe (2007)	Shamir & Kark (2004)	Included
Scales similar to organizational commitment scales	Balfour & Wechsler (1991)	Cheney (1983)	Excluded
	Gautam et al. (2004)	Cheney (1983)	Excluded
	Ishii (2012)	Cheney (1983)	Excluded
	Johnson et al. (1996)	Cheney (1983)	Excluded
	Sass & Canary (1991)	Cheney (1983)	Excluded
	Scott et al. (1999)	Cheney (1983)	Excluded
	van Dick et al. (2006) Sample 10	Cheney (1983)	Excluded
	Wolf (2009)	Cheney (1983)	Excluded
	Gould & Werbel (1983)	Patchen (1970)	Excluded
	Popoola (2005)	Patchen (1970)	Excluded
	Rotondi (1975)	Patchen (1970)	Excluded
	Jetten et al. (2002)	Allen & Meyer (1996)	Excluded
	Olkkonen & Lipponen (2006)	Allen & Meyer (1990)	Excluded
	O'Reilly & Chatman (1986)	O'Reilly & Chatman (1986)	Excluded
	Leavitt et al. (2011)	O'Reilly & Chatman (1986)	Excluded
	Kolodinsky et al. (2008)	Efraty et al. (1991)	Excluded
	Millward & Brewerton (1999)	Millward & Brewerton (1999)	Excluded
	Tyagi & Wotruba (1993)	Not available	Excluded

agreement rates were high, ranging from 98% to 100% for the coding of job involvement, job satisfaction, affective organizational commitment, extra-role performance, national culture, data structure, and publication status. The initial intercoder agreement was somewhat lower for the coding of in-role performance (Cohen's $\kappa = .92, p < .001$). Disagreements between raters were resolved by discussing coding criteria and further examination of the studies, which eventually resulted in an intercoder agreement rate of 100% for all coded information.

Outcomes of organizational identification. Two categories were used to code the outcomes of organizational identification: attitudinal and behavioral outcomes. *Attitudinal outcomes* were coded using three subcategories: job involvement, job satisfaction, and affective organizational commitment. *Behavioral outcomes* were coded using two subcategories: in-role performance and extra-role performance. Some studies measured the constructs of in-role performance and extra-role performance using several dif-

ferent variable names. In such cases, we classified the variables into in-role and extra-role behaviors based on their definitions; performance behaviors were classified into in-role performance when they were required by formal job descriptions, directly serving the goals of the organization (van Knippenberg, 2000), while performance behaviors were classified into extra-role performance when they concerned discretionary actions beyond formal job descriptions (Podsakoff et al., 2000). For example, we coded variables such as productivity, job performance, customer-oriented service behavior, and work effort into in-role performance, and coded variables such as organizational citizenship behavior, helping behavior, voice behavior, and safety performance into extra-role performance. We report all of our classification information in the Appendix.

In addition, we coded the rater of performance variables because several researchers noted that self-reported measures can be affected by the self-serving bias (e.g., Borman, 1991; Spector, 1994).

Table 2
Stem and Leaf Display of 114 Correlations (*r*)

Stem	Leaf
-.1	
-.0	5,4
.0	1,2,3,3,3,4,5,5,5,5,6,8,8,9,9,9
.1	1,2,3,4,5,5,6,8,8,8,9,9
.2	0,0,0,0,0,0,1,2,2,2,3,3,3,4,4,4,5,5,5,6,6,7,7,7,8,8,9,9,9,9,9
.3	0,0,1,1,1,1,2,2,3,3,3,3,3,4,4,4,5,6,6,6,6,6,7,7,8,8,9,9,9
.4	0,0,0,1,1,1,1,1,2,3,3,3,3,4,4,5,6,6,6,7,7,8,8,8,8,9,9
.5	0,0,0,0,1,1,1,2,2,2,3,3,3,4,5,6,6,7,7,8,8,8
.6	1,3,5,6,6
.7	3
.8	

We distinguished in-role performance outcomes into self-reported, other-reported, and data-based. Similarly, we distinguished extra-role performance outcomes into self-reported and other-reported (Allen, Barnard, Rush, & Russell, 2000; Carpenter, Berry, & Houston, 2014). No study in our sample measured extra-role performance based on objective data. Moreover, we also distinguished extra-role performance by target: organizational citizenship behavior toward individuals (OCB-I) and toward the organization (OCB-O), following the previous notion of the distinctive characteristics of the two (Podsakoff, Whiting, Podsakoff, & Blume, 2009; Williams & Anderson, 1991).

National culture. We coded *national culture* using the values provided by Hofstede, Hofstede, and Minkov (2010), as de Wit, Greer, and Jehn (2012) did in their meta-analysis. We first identified the geographical location in which the study was conducted. Example countries included in our study are the United States, Canada, Germany, China, France, the Netherlands, the Philippines, Thailand, Singapore, Pakistan, Australia, and Japan, to list a few (see the Appendix for all of the country information). After identifying the geographical locations, we assigned to each study the associated values of Hofstede et al.'s (2010) three cultural value dimensions: individualism/collectivism, long-term/short-term orientation, and uncertainty-avoidance. For example, the United States was coded 91 for individualism, 29 for long-term-orientation, and 46 for uncertainty-avoidance. The same dimensions were coded 20, 118, and 30, respectively, for studies conducted in China.

Study characteristics. We coded two aspects of study characteristics for additional exploratory moderator analyses. First, several researchers (e.g., Judge, Thoresen, Bono, & Patton, 2001) have noted that data structure, for example cross-sectional versus longitudinal, can influence correlations; cross-sectional data may present higher correlations than longitudinal data because of common method variance (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Thus, we coded *data structure* into three types: cross-sectional data (concurrent measures of organizational identification and its outcomes), longitudinal data (time separation between the measurement of organizational identification and its outcomes), and panel data (correlations between average organizational identification across times and average outcomes across times). Second, the issue of publication bias, defined as the selective publication of studies that have statistically significant findings, has been an important issue for psychological science for at

least three decades (e.g., Ferguson & Brannick, 2012; Rosenthal, 1979, 1995). To investigate whether publication bias exists in the organizational identification literature, we coded *publication status* into two categories: studies published in academic journals versus unpublished manuscripts and dissertations.

Meta-Analysis Procedure

All the correlations were first corrected for measurement error in the independent and dependent variables. We used the internal consistency coefficients reported in the respective study as the reliability estimates. When reliability estimates were available, we divided individual effect sizes by the square root of the reliability estimates of the two correlated variables, as recommended by Hunter and Schmidt (2004). Among the initial sample of 149 correlations, 12 correlations (8%) did not have the reliability estimate information for the organizational identification variable and 36 correlations (24%) did not have the reliability estimate information for the outcome variables. For studies that did not report a reliability estimate, we assigned them the average reliability estimate based on the estimates from other studies in our sample (Hunter & Schmidt, 2004). The artifact distribution of the reliability of organizational identification was a mean of .82 with a standard deviation of .07. The artifact distributions of the criterion variables' reliabilities were mean .77 and standard deviation .11 for job involvement; mean .80 and standard deviation .09 for job satisfaction; mean .83 and standard deviation .05 for affective organizational commitment; mean .77 and standard deviation .11 for in-role performance; and mean .79 and standard deviation .09 for extra-role performance.

We calculated meta-analytic correlations using a random effects model (Hunter & Schmidt, 2004) to consider heterogeneity among the studies. The random effects model assumes that sampling error plus variability in the population of the correlations (unique differences in the set of true population correlations) caused the variability among the organizational identification–outcome correlations. As an estimate of variability, we calculated 95% confidence intervals and 80% credibility intervals. A 95% confidence interval excluding zero indicates that one can be 95% confident

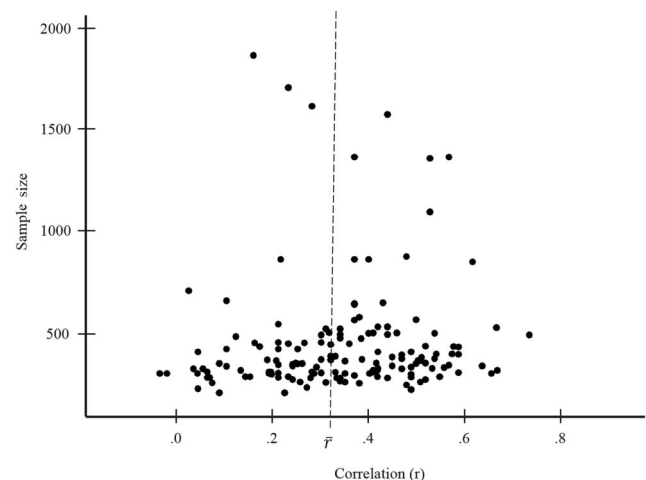


Figure 3. Funnel plot of 114 independent correlations (*r*).

that the average true correlation is different from zero; thus, a maximum of 2.5% are larger than the upper bound of the interval and fewer than 2.5% are smaller than the lower bound. A credibility interval indicates whether the correlations reported are generalizable to other samples. For example, an 80% credibility interval excluding zero indicates that at least 80% of the correlations reported are different from zero. In addition, we conducted homogeneity analysis (Lipsey & Wilson, 2001), which tests whether it is reasonable to assume that all effect sizes are estimating the same population mean, using two indices: the Q -statistic, which indicates the level of variance across study results relative to the sampling error variance (Hedges & Olkin, 1985), and the I -squared (I^2) statistic, another indicator of homogeneity statistic (Higgins, Thompson, Deeks, & Altman, 2003). The Q -statistic indicates whether a statistically significant level of variability exists in correlations across studies, by calculating the categorical model results for the between-groups goodness-of-fit statistic (Q_B) and the within-group goodness-of-fit statistic (Q_W ; Field, 2001; Hedges & Olkin, 1985; Lipsey & Wilson, 2001). The I^2 indicates the ratio of true heterogeneity to total variation in observed effect sizes, which ranges from 0% to 100% with higher values indicating greater heterogeneity of effect sizes and higher likelihood of moderators. Compared with the Q -statistic, the I^2 is known to be less affected by the scaling of the measures or the number of the studies included (Borenstein, Hedges, Higgins, & Rothstein, 2009). However, the I^2 should also be interpreted with caution because it depends on the size of individual studies (Higgins & Thompson, 2002).

Finally, we created a correlation matrix using the meta-analytic correlations and conducted fixed-effect meta-analytic structural equation modeling (MASEM; Viswesvaran & Ones, 1995) to test our hypothesized path models (in Figures 1 and 2). In addition, to test the moderating effect of national culture on the relations between organizational identification and its outcomes, we conducted a random-effects weighted least square (WLS) regression analysis using the meta-regression SPSS syntax developed by Lipsey and Wilson (2001).

Results

Relations Between Organizational Identification and Outcomes

Table 3 presents the analysis of the relations between organizational identification and its attitudinal/behavioral outcomes using the available correlations.

The top panel of Table 3 reports the meta-analysis results using all available independent correlations ($k = 114$; $N = 36,526$). The average corrected correlation between organizational identification and all attitudinal/behavioral outcomes across all studies was positive ($\hat{\rho} = .41$), and the 95% confidence interval did not include zero (95% CI [.37, .45]). However, the corrected correlation showed large variance; the homogeneity of effect size tests was significant across the analyses ($Q_W = 1587.13$, $p < .01$). This indicates that the use of a random effects model is justifiable. It also indicates that moderators may be present for the associations between organizational identification and outcomes.

The second panel of Table 3 presents the tests of the correlations between organizational identification and outcomes by outcome type: attitudinal outcomes and behavioral outcomes. The between-groups goodness-of-fit statistic Q_B showed that the correlations between organizational identification and outcomes were significantly different across outcome types ($Q_B(1) = 17.96$, $p < .01$). Specifically, the results revealed that the size of the correlation between organizational identification and behavioral outcomes ($\hat{\rho} = .35$, 95% CI [.32, .39]) was smaller than that between organizational identification and attitudinal outcomes ($\hat{\rho} = .49$, 95% CI [.44, .54]).

The third and fourth panels of Table 3 present the organizational identification correlations with detailed outcomes within each outcome type. The third set of results in Table 3 shows that organizational identification was significantly and positively related to job involvement ($\hat{\rho} = .50$, 95% CI [.39, .62]), job satisfaction ($\hat{\rho} = .45$, 95% CI [.40, .50]), and affective organizational commitment ($\hat{\rho} = .64$, 95% CI [.55, .72]).

The fourth set of results in Table 3 reports the tests of the relations between organizational identification and behavioral outcomes. The Q_B indicates that the correlations between organizational identification and behavioral outcomes were significantly different across performance types ($Q_B(1) = 19.64$, $p < .01$). Specifically, the correlation between organizational identification and in-role performance was significant and positive ($\hat{\rho} = .27$, 95% CI [.20, .34]) and was smaller than the correlation between organizational identification and extra-role performance ($\hat{\rho} = .42$, 95% CI [.36, .48]). We further classified both performance outcomes based on the rater of performance. The relation between organizational identification and in-role performance was relatively high when in-role performance was self-rated ($\hat{\rho} = .33$, 95% CI [.23, .38]) and low when in-role performance was rated based on objective data ($\hat{\rho} = .19$, 95% CI [.03, .34]). The relation between organizational identification and extra-role performance was higher when extra-role performance was measured using self-rated measures ($\hat{\rho} = .48$, 95% CI [.43, .52]) than when it was measured using other-rated measures ($\hat{\rho} = .29$, 95% CI [.18, .39]) and the difference between the two was statistically significant ($Q_B(1) = 10.38$, $p < .01$). In addition, we also examined whether the relation between organizational identification and extra-role performance differed by target—OCB-I and OCB-O—and the results showed that they were not significantly different ($Q_B(1) = 2.02$, ns).

The fifth panel of Table 3 shows how the organizational identification–outcome correlations change depending on data structure—the study design used to collect the data. The results indicated that data structure did not significantly moderate the relations between organizational identification and its outcomes ($Q_B(2) = 2.09$, ns).

Positioning of Organizational Identification in General Attitude–Behavior Relations

We used MASEM (Viswesvaran & Ones, 1995) to test how organizational identification functions in the nomological network of attitudinal and behavioral variables. To conduct the MASEM analysis, we first constructed the matrix of meta-

Table 3

Meta-Analysis of Bivariate Relations Between Organizational Identification and Outcome Variables

Variable	<i>k</i>	<i>N</i>	$\bar{r}$	SD_r	$\hat{\rho}$	SD_{ρ}	95% CI	80% CV	I^2	Q_B	Q_W
All outcomes											
All independent correlations	114	36,526	.34	.21	.41	.21	[.37, .45]	[.14, .69]	93%		1587.13**
Outcome type										17.96**	150.78
Attitudinal outcomes	55	18,084	.41	.14	.49	.22	[.44, .54]	[.25, .73]	0%		46.90
Behavioral outcomes	94	43,754	.29	.19	.35	.19	[.32, .39]	[.11, .60]	10%		103.88
Attitudinal outcomes										13.97**	55.71
Job involvement	6	3,939	.41	.12	.50	.15	[.39, .62]	[.31, .69]	42%		8.68
Job satisfaction	37	11,216	.37	.12	.45	.18	[.40, .50]	[.22, .68]	0%		32.25
Affective organizational commitment	12	2,929	.52	.14	.64	.17	[.55, .72]	[.42, .86]	26%		14.77
Behavioral outcomes										19.64**	94.29
In-role performance	41	13,870	.22	.13	.27	.26	[.20, .34]	[−.06, .60]	27%		39.78
In-role performance by rater										2.66	40.43
Self-rated	20	8,468	.27	.13	.33	.18	[.23, .38]	[.13, .53]	14%		22.16
Other-rated	16	3,315	.18	.16	.23	.16	[.14, .31]	[.03, .43]	5%		15.83
Data-based	5	2,087	.09	.13	.19	.18	[.03, .34]	[−.04, .42]	0%		2.44
Extra-role performance	53	14,459	.35	.15	.42	.22	[.36, .48]	[.18, .66]	5%		54.52
Extra-role performance by rater										10.38**	48.75
Self-rated	39	9,928	.38	.12	.48	.19	[.43, .52]	[.27, .68]	0%		34.66
Other-rated	9	1,753	.24	.12	.29	.15	[.18, .39]	[.08, .49]	21%		14.09
Extra-role performance by target										2.02	22.94
OCB-I	5	927	.23	.18	.27	.20	[.09, .45]	[.01, .53]	0%		1.34
OCB-O	18	4,089	.34	.17	.42	.21	[.32, .51]	[.15, .69]	21%		21.60
Study characteristics											
Data structure										2.09	149.41
Cross-sectional	138	59,419	.34	.12	.41	.23	[.38, .45]	[.16, .67]	0%		140.33
Longitudinal	10	2,275	.26	.16	.32	.19	[.19, .44]	[.05, .58]	0%		9.08
Panel ^a	1	144	.36	.17	.39	.21	[−.01, .79]	[.13, .65]	—		—
Publication status										.70	149.67
Academic journal articles	140	44,188	.34	.12	.41	.24	[.38, .44]	[.15, .67]	3%		143.56
Unpublished manuscripts/dissertations	9	2,225	.29	.18	.35	.21	[.22, .48]	[.09, .61]	0%		6.12

Note. *k* = number of correlations from independent samples; *N* = total sample size for all studies combined; $\bar{r}$ = sample-size weighted mean correlation; SD_r = sample-size weighted observed standard deviation of correlation; $\hat{\rho}$ = mean true-score correlation corrected for measurement error in both predictor and criterion measures; SD_{ρ} = standard deviation of the true-score correlation corrected for measurement error in both predictor and criterion measures; 95% CI = lower and upper limits of 95% confidence interval; 80% CV = lower and upper limits of 80% credibility interval; I^2 = homogeneity statistic; Q_B = homogeneity statistic *Q* between groups; Q_W = homogeneity statistic *Q* within group; OCB-I = organizational citizenship behavior toward individuals; OCB-O = organizational citizenship behavior toward the organization.

^a The meta-analytic correlation in this row may not be meaningful because *k* is less than 3, but we decided to report this for informational purposes.

** $p < .01$.

analytic correlations among all of the study variables (see Table 4). We then entered this meta-analytic correlation matrix into an SEM analysis using AMOS (Version 18; Arbuckle, 2009). As the sample sizes across the various cells of the matrix were not identical, we calculated the harmonic mean of the sample sizes and used it for the SEM analyses (Viswesvaran & Ones, 1995). The harmonic mean is a more conservative parameter estimate than the arithmetic mean because it assigns much less weight to large sample sizes. We used five model fit indices—chi-square (χ^2), the comparative fit index (CFI), the Tucker-Lewis index (TLI; also known as the non-normed fit index, Tucker & Lewis, 1973), the root-mean-square error of approximation (RMSEA), and the standardized root-mean-square residual (SRMR)—to investigate the viability of the structural models. Desirable model fit is associated with a CFI greater than .90, a TLI greater than .90, an RMSEA less than or equal to .08, and an SRMR less than .10 (Kline, 2005).

Table 5 presents the comparison of two alternative models that position organizational identification differently in the attitude-behavior relations. First, we tested the model of *organizational*

identification as a component of general attitude that treats organizational identification as constituting a general set of attitudes along with other types of attitudes (job involvement, job satisfaction, and affective organizational commitment; Model 1 in Figure 1). The MASEM results show that Model 1 did not fit the data particularly well ($\chi^2[8] = 647.42$, CFI = .95, TLI = .90, RMSEA = .11, SRMR = .04). Next, we tested the model of *organizational identification as a basis for general attitude and behavior* that treats organizational identification not as a part of general attitude but as an independent predictor of both general attitude and general behavior (Model 2 in Figure 2). When Model 2 was estimated, all of the fit indices reached acceptable levels ($\chi^2[7] = 344.21$, CFI = .97, TLI = .94, RMSEA = .08, SRMR = .03). In addition, to further compare the two models, we utilized Akaike's information criterion (AIC; Akaike, 1974), which is generally used in SEM to compare non-nested models estimated using the same data (Henson, Reise, & Kim, 2007; Kline, 2005). The AIC for Model 2 (372.21) was smaller than that for Model 1 (673.42). As a lower AIC value indicates a better fit, this result indicates that Model 2 fit the data better than Model 1.

Table 4
Meta-Analytic Correlations ($\hat{\rho}$) Between Study Variables

Variable	1	2	3	4	5
1. Organizational identification					
2. Job involvement	.50 ^a				
<i>k</i> studies	6				
<i>N</i> total observations	3,939				
3. Job satisfaction	.45 ^a	.35 ^b			
<i>k</i> studies	37	462			
<i>N</i> total observations	11,216	133,062			
4. Affective organizational commitment	.64 ^a	.50 ^b	.60 ^d		
<i>k</i> studies	12	22	112		
<i>N</i> total observations	2,929	6,100	39,187		
5. In-role performance	.27 ^a	.18 ^b	.23 ^c	.22 ^f	
<i>k</i> studies	41	85	312	14	
<i>N</i> total observations	13,870	21,618	54,471	3,741	
6. Extra-role performance	.42 ^a	.30 ^c	.24 ^c	.29 ^f	.23 ^d
<i>k</i> studies	53	10	22	14	24
<i>N</i> total observations	14,459	3,612	7,100	3,711	9,912

Note. The letter superscripts in the table indicate the sources of the meta-analytic correlations as follows: ^a Our meta-analysis. ^b Cooper-Hakim & Viswesvaran (2005). ^c Newman et al. (2010). ^d Harrison et al. (2006). ^e LePine et al. (2002). ^f Colquitt et al. (2007).

Figure 4 presents the standardized path estimates for the supported model (Model 2). Organizational identification was positively and significantly related to both general attitude ($\beta = .74$, $p < .001$) and general behavior ($\beta = .53$, $p < .001$). Its effect on general behavior was even stronger than the effect of general attitude on general behavior ($\beta = .24$, $p < .001$). Overall, the results show that the attitude-behavior model fit the data better when organizational identification was specified as an independent construct—not as a component of general attitude—and organizational identification was a strong predictor of both general attitude and general behavior in organizations.

Moderation of National Culture

To test how national culture moderates the effects of organizational identification on attitudes/behaviors, we ran a WLS regression model by using organizational identification–outcome correlations as a dependent variable, and the results are reported in Table 6. As shown in Table 6, among the three cultural dimensions, the individualism dimension had significantly negative effects on the strength of the relations between organizational identification and outcomes ($\beta = -.20$, $p < .05$), meaning that the organizational identification–outcome relations were stronger in collectivistic cultures than in individualistic cultures. Other dimensions such as long-term-orientation (i.e., time-orientation; $\beta = .11$, ns) and uncertainty-avoidance ($\beta = .09$, ns) did not have significant effects on the strength of the relations between organizational identification and outcomes.

Publication Bias Analysis

Following the standard practice for reporting meta-analysis results, we evaluated potential publication bias, which typically occurs when the publicly available literature is not representative of all studies on the relation of interest (Rothstein, Sutton, & Borenstein, 2005). Specifically, we tested for publication bias in the following three ways. First, we analyzed our data using publication status (academic journal articles vs. unpublished manuscripts and dissertations) as a moderator of the relations between organizational identification and its outcomes. As reported in the last panel of Table 3, the test results show that the relations between organizational identification and outcomes did not depend on publication status ($Q_B(1) = .70$, ns); the average correlation reported in academic journal articles ($\hat{\rho} = .41$) was not significantly different from the average correlation reported in unpublished manuscripts and dissertations ($\hat{\rho} = .35$). Second, we tested the statistical significance of the correlation between the sample size and the organizational identification–outcome correlation to investigate whether the correlation of each study is related to its sampling error. The results show that neither the correlation between sample size and organizational identification–outcome correlation ($r = .05$, ns) nor the correlation between organizational identification–outcome correlation and its standard error ($r = -.05$, ns) was significant. These results indicate that the correlation sizes were not related to the precision of the studies' measurements. Third, we examined the magnitude and the direc-

Table 5
MASEM Results: Positioning of Organizational Identification in General Attitude–Behavior Relations

Model	χ^2	<i>df</i>	CFI	TLI	RMSEA	SRMR	AIC
Model 1: Organizational identification as a component of general attitude	647.42	8	.95	.90	.11	.04	673.42
Model 2: Organizational identification as a basis for general attitude and behavior	344.21	7	.97	.94	.08	.03	372.21

Note. Harmonic mean $N = 7,001$.

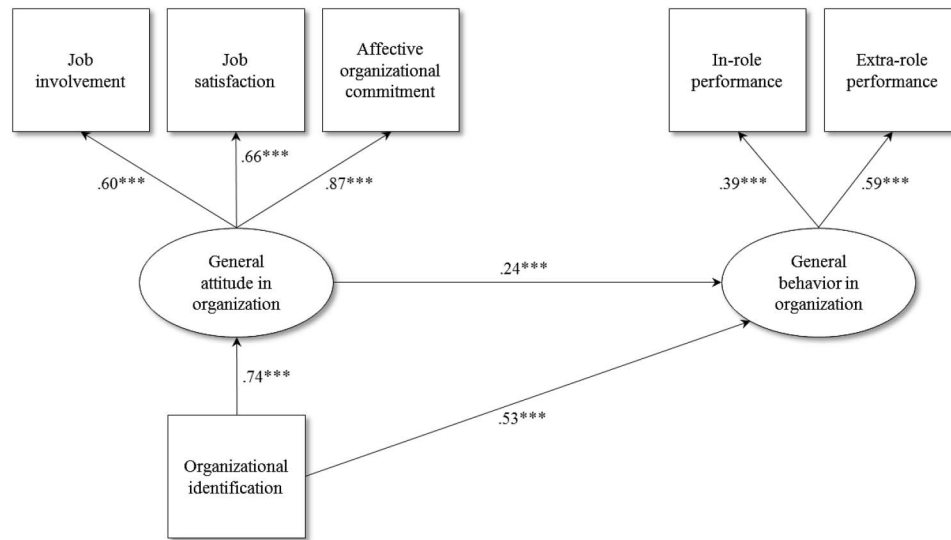


Figure 4. Final results of the positioning of organizational identification in attitude-behavior relations. Harmonic mean $N = 7,001$. Standardized coefficients are presented. *** $p < .001$.

tion of publication bias using meta-regression analysis (Stanley, 2008). Specifically, we conducted a weighted regression, weighted by the inverse of the standard error, by regressing the correlation of each study on its standard error. The result was not significant ($\beta = -2.80$, ns), which indicates that publication bias is unlikely to be present.

Discussion

Being a member of and working for an organization constitute a substantial part of one's life, and individuals often construct their self-concepts from the identity of the organiza-

tion they belong to. In this vein, organizational identification has often been regarded as a "root construct" (Albert et al., 2000, p. 13) which provides a basis on which individual attitudes and behaviors are developed. Using 149 correlations from 114 studies on organizational identification, we aimed to provide meta-analytic evidence that reveals the unique value of this identity-relevant construct. Our results clarified the theoretical positioning of organizational identification that underlies work attitudes and behaviors in two ways; we demonstrated that (a) organizational identification was significantly associated with all the key attitudinal/behavioral constructs in organizations, and (b) organizational identification, in the nexus of attitude-behavior relations in organizations, has unique effects on behaviors, above and beyond the effects that attitudes have on them. Furthermore, we explored how these effects of organizational identification on attitudes/behaviors are moderated by national culture—an important higher-level social context wherein the organization is embedded—and revealed that the individualism/collectivism cultural dimension substantially moderates the organizational identification effects. Below, we discuss each finding and its implications from both theoretical and practical standpoints.

Organizational Identification as an Antecedent of Work Outcomes

We expected organizational identification to significantly influence the key organizational constructs because organizational identification provides a positive subtext for individuals developing attitudes and behaviors favorable to organizations. Consistent with our expectation, the results of our meta-analysis reveal that organizational identification is positively and significantly correlated with attitudinal outcomes—job involvement, job satisfaction, and affective organizational commitment—and behavioral outcomes—in-role performance and extra-role performance.

Table 6

Meta-Regression Analysis Results: The Effects of National Culture on the Organizational Identification–Outcome Relations

Cultural dimension	β	z	p
Intercept	.00	8.99	.00
Individualism ^a	-.20	-2.50	.01
R^2	.04		.01
v [se(v)]	.04 [.00]		
Intercept	.00	9.40	.00
Long-term-orientation	.11	1.33	.18
R^2	.01		.18
v [se(v)]	.04 [.00]		
Intercept	.00	5.76	.00
Uncertainty-avoidance	.09	1.12	.26
R^2	.01		.26
v [se(v)]	.04 [.00]		

Note. $k = 146$. k includes all attitudinal and behavioral outcomes. Analyses were conducted using a weighted least square regression model. β = standardized regression coefficient; z = critical ratio test for the regression coefficient; p = two-tailed probability of z (i.e., p -value); v = maximum likelihood random effects variance component; se(v) = standard error of estimated variance component.

^a High individualism score indicates low collectivism.

In addition to the statistically significant associations, the results also indicate that the strength of the associations varies by the type of work outcome. Regarding attitudinal outcomes, we found that among the three work attitudes, affective organizational commitment was the most strongly associated with organizational identification. This implies that although organizational identification meaningfully generates all types of positive work attitudes, its effects become more salient when the target of attitude is one's organization than when it is one's job.

With respect to behavioral outcomes, a notable finding is that organizational identification was more strongly associated with extra-role performance ($\hat{\rho} = .42$) than with in-role performance ($\hat{\rho} = .27$), implying that organizational identification's unique identity-relevant nature—making individuals perceive that acting on behalf of the organization is tantamount to acting on behalf of themselves—is more obviously expressed in extra-role performance. Individuals with high organizational identification are more likely to “go the extra mile” voluntarily, performing tasks that may primarily benefit the whole organization rather than themselves. Therefore, organizational identification has more prominent effects on discretionary, nonrewarded performance beyond formal job requirements (i.e., extra-role performance) than on individually rewarded job performance (i.e., in-role performance). Another possible explanation for organizational identification's stronger association with extra-role performance may be that in-role performance is affected by many other factors beyond the individual's direct control, while extra-role performance is basically at the individual's discretion (van Dick et al., 2006). As an additional analysis, we further distinguished extra-role performance with respect to its targets—individual (OCB-I) and organization (OCB-O)—but their correlations with organizational identification were not significantly different from one another. The result implies that individuals who strongly identify with their organizations exhibit high extra-role performance regardless of its targets. From the perspective of social exchange (Blau, 1964), OCB-I has more potential for reciprocal benefits than, thus preferred to, OCB-O because OCB-O, with relative lack of a specific target of behavior, is more likely to be unnoticed or unperceived by other employees in the organization (Johnson & Chang, 2006). In case of strong identifiers, however, as they perceive the organization as a whole with the oneness perception between themselves and the organization, they are likely to engage in OCBs not specifically differentiating the targets and their relevant reciprocal benefits. Taken together, our meta-analysis results regarding behavioral outcomes consistently indicate that when engaging in certain work behaviors, individuals with high organizational identification are not likely to make distinction between self-interest and the organization's interest; they are just ready to go the extra mile.

Organizational Identification's Unique Value as a Basis for Attitudes and Behaviors

The results of our study offer important theoretical implications for the classic psychological model of attitude-behavior relations by identifying the positioning of organizational identification in the nexus of attitudes/behaviors in organizations. Advancing the

previous test of the compatibility principle in organizational contexts (Harrison et al., 2006; Newman et al., 2010), we brought organizational identification into the general attitude-behavior framework, specifying two alternative theoretical models. In comparing the model in which organizational identification serves as a component of general attitude and the model in which organizational identification serves as an independent construct that shapes general attitude and behavior, we found that the latter model fit the data better than the former. These results indicate that organizational identification, as a unique construct separate from other attitude constructs, functions as a fundamental basis for other attitudes and behaviors developed in organizations.

The size of the coefficients in the finally selected structural path model further demonstrates the unique value of organizational identification. In predicting general behavior, the direct path from organizational identification ($\beta = .53$) was even stronger than that from the general set of attitudes ($\beta = .24$). This indicates that organizational identification not only explains a significant amount of variance in general behavior above and beyond what is explained by general attitude, but it also explains *more* variance than general attitude does. This strong explanatory power of organizational identification may be present because it reflects an essential “visceral connection” (Ashforth et al., 2008, p. 333) between individual and organizational identities—other attitude constructs such as organizational commitment lack that depth of connection (Ashforth et al., 2008). Thus, an organizationally identified individual, as a microcosm of the organization, behaves in the organization's best interests. Overall, our findings reveal that it is indeed important for organizational behavior and effectiveness to enhance employees' essential sense of “who I am” and “who we are” as a member of the organization.

A notable underlying theoretical implication of this finding involves the relative motivating power of *identity-based* concerns versus *interest-based* concerns (Blader & Tyler, 2009; Haslam & Ellemers, 2005; Kelly & Kelly, 1994; Tyler & Blader, 2000, 2003). As discussed above, affective organizational commitment explains the individual-organization relationship in terms of social exchange between separate psychological entities. Individuals with high affective organizational commitment are more job-oriented than those with low commitment because they view their jobs as fulfilling more of their individual interests in the exchange relationship with the organization (Bateman & Strasser, 1984; Van Scotter, 2000; Vandenberg & Lance, 1992). This implies that affective organizational commitment is substantially related to whether the job or work itself is satisfying, meaningful, challenging, and not emotionally exhausting (Mathieu & Zajac, 1990; Meyer & Allen, 1997; Meyer, Stanley, Herscovitch, & Topolnysky, 2002). Thus, attitude constructs—job involvement, job satisfaction, and affective organizational commitment—all involve individual *interests* such as social/economic exchange for benefits like support, recognition, and pay, which constitute important resources of job and organizational experiences (Blau, 1964; Eisenberger, Huntington, Hutchison, & Sowa, 1986). Indeed, Pratt (1998) argued that commitment is similar to job satisfaction and related to the question, “How happy or satisfied am I with the organization?” (p. 178). Pratt (1998) also suggested that organizational identification, by contrast, is basically associated with, “How do I perceive myself in relation to my organization?” (p. 178), which denotes that organizational identification captures

more fundamental concerns regarding *identity*—psychological oneness between the individual and the organization. Thus, our meta-analysis finding that organizational identification, as an identity-relevant construct, explains performance outcomes (in-role and extra-role performance) more than interest-based attitudes do reveals that the resources (e.g., pay, support) at work are not necessarily the primary motivators that generate desirable behaviors in organizations. Rather, desirable behaviors in organizations are substantially engendered when individuals integrate the contextualized social identity (i.e., organizational membership) that gives those resources “meaning” (Haslam & Ellemers, 2005, p. 65) into their self-concept, thus becoming intrinsically motivated (van Knippenberg, 2000).

Taken together, these discussions confirm that organizational identification is essentially different from work attitude constructs. More importantly, to improve employee performance, organizations need *identified* individuals (those having intrinsic psychological oneness), beyond *committed* or *satisfied* individuals (those having attitudes based on the notion of instrumental exchange). Further, as evidenced in our result revealing the significant path from organizational identification to general attitude, when individuals are rooted in the organization (i.e., identified with the organization), their commitment or satisfaction may follow.

An important practical implication of our findings is that organizations can benefit more from identity-enhancing programs than from interest-enhancing programs. Programs designed to enhance employees’ sense of interest—for example, providing exchange-based, instrumental motivators such as a performance-based reward system or making the job/working conditions more enjoyable and pleasant—may help enhance employees’ work attitudes and subsequent performance, to some extent. However, those programs do not necessarily improve employees’ organizational identification, which constitutes a basis for both attitudes and behaviors. Instead, by making more active use of programs designed to enhance employees’ sense of identity as an organizational member, such as organizational socialization tactics (Ashforth & Saks, 1996; Van Maanen & Schein, 1979) and internal branding programs focusing on organizational prestige and distinctiveness (Dutton et al., 1994), organizations may maximize the benefits from employees’ organizational identification. The effects of such identity-enhancing programs will be longer-term compared with interest-enhancing programs and will function as intrinsic motivators beyond instrumental motivators.

Despite those theoretical and practical implications, our findings should be interpreted with caution. In particular, we acknowledge that our model may only apply to the bright side of organizational phenomena—organizationally desirable attitudes/behaviors. Due to the lack of available empirical studies, we could not meta-analyze the relations between organizational identification and the behavioral outcomes reflecting the dark side of organizational life (the number of studies was 1 for actual turnover, 1 for absenteeism, 0 for lateness, and 3 for counterproductive work behaviors). The scarcity of such studies precluded us from including, in our models, negative behavior as another potentially important factor of general behavior in addition to the favorable outcomes of in-role and extra-role performance (Harrison et al., 2006; Newman et al., 2010), thereby reducing the generalizability of our findings. We thus expect future research to explore organizational identification’s implications for those undesirable behaviors at work, to

present a more general picture of the subtext that organizational identification provides for a more comprehensive set of attitudes/behaviors.

Another potential extension of our study would be to explore other possibilities of causal directions among variables. Regarding attitude-behavior relations, self-perception theory (Bem, 1965, 1972) postulates reverse causality such that behavior causes attitude; individuals often develop their attitudes from the implications of their own past behavior and the circumstances in which that behavior occurs (Albarracín & Wyer, 2000). Similarly, researchers have also noted reverse causality with respect to identity. They suggested that individuals learn about themselves by observing their own behaviors (Chang et al., 1988; Markus, 1977; Sparks & Shepherd, 1992); repeated past behaviors create, affirm, or bolster a person’s self-identity because “carrying out these acts conveys meaning over and above the positive or negative attitudes we may hold toward performing the behavior itself” (Chang et al., 1988, p. 304) and that meaning is ultimately incorporated into the self-concept, forming “self-schemata”—a cognitive structure about the self (Markus, 1977). Thus, in organizational settings, following repeated performance of behaviors such as in-role and extra-role performance, self-identity as an organizational member will become more salient. However, we could not examine this notion because the studies included in our meta-analysis consistently conceptualized behavior as a consequence of organizational identification, and none of them examined the organizational identification-behavior relations while considering reverse causality, particularly using a longitudinal study design. As MASEM is based on the data that compose the meta-analytic correlation matrix, causal interpretations should be made with caution, unless the primary studies from which those correlations are obtained permit certain causal assertions (Landis, 2013). Our meta-analysis was not based on the primary studies that investigated reverse causality, and hence it was not possible to provide meta-analytic evidence of reverse causality between organizational identification and behaviors. Therefore, we will leave it as a role of future researchers to explore the causal direction of the relation between organizational identification and behavior in organizational settings by collecting data at multiple points in time.

Organizational Identification Effects Moderated by National Culture

We presented the first systematic empirical evidence of how national culture moderates the effects of organizational identification on attitudinal/behavioral outcomes. The results of our meta-analysis reveal that the organizational identification effects were stronger in collectivistic than in individualistic cultures. This implies that identification with the collective (i.e., organization) has more salient effects on attitudinal/behavioral outcomes when the individuals are situated within the value system of a collectivistic national culture that congruently emphasizes the self-concept as constructed in a collective manner. From a practical perspective, our results also indicate that the aforementioned organizational programs intended to enhance employees’ organizational identification can be more effectively used in collectivistic cultures than in individualistic cultures.

Our finding, however, does not indicate that the organizational identification effects disappear in an individualistic cul-

tural context. Back-of-the-envelope calculation of our data showed that the size of the organizational identification effect was .55 when the individualism score was the lowest (14) and it was .37 when the individualism score was the highest (99). That is, the size of the organizational identification effect remained much greater than zero even when the national culture is highly individualistic. This demonstrates that, although its effects are not as strong as in collectivistic cultures, organizational identification remains an important root that substantially shapes attitudes and behaviors even in a highly individualistic culture.

We did not find significant statistical evidence that the other two dimensions—time-orientation and uncertainty-avoidance—influence the organizational identification–outcome relation. Nevertheless, we believe that the insignificant meta-analysis results do not necessarily deny the possibility that the two cultural dimensions have explanatory power for the effects of organizational identification. In our meta-analysis, we assigned the associated values of Hofstede et al.'s (2010) cultural dimensions to each study, assuming that all individuals in the specific nation share the same perceptions of their nation's cultural characteristics. This approach has been widely used and is known as acceptable (see Chen, Leung, & Chen, 2009 and Tsui, Nifadkar, & Ou, 2007 for comprehensive discussions on this issue), but some researchers have suggested that cultural effects on work attitudes/behaviors vary within nations because of significant individual differences in beliefs about and internalization of their national culture's value system (Brockner, 2003; Meglino & Ravlin, 1998). This implies that the organizational identification–outcome relation may also be considerably influenced by within-nation individual differences. Therefore, the nation-level (i.e., study-level) measures of culture used in our analysis might not have appropriately captured the cultural effects on the organizational identification–outcome relation, thus generating less significant results. Future research on cultural differences in organizational identification effects would thus benefit from further investigation of culture as measured by individual perceptions, not as measured by nation-level dimensions.

Conclusion

We began this article with the notion that every entity needs an essential sense of “Who am I?” or “Who are we?” to develop attitudes and behaviors adequate to the surrounding social contexts, particularly organizations. The importance of identity and identification in organizational life has been emphasized by many scholars for the past decades, but systematic evidence of how organizational identification is positioned in the nexus of attitudes/behaviors in organizations was scarce. We believe that our study contributes to advancing identity-related arguments and discussions by providing meta-analytic evidence of the unique value of organizational identification; the sense of identity captured by this construct essentially contributes to shaping behaviors favorable to the organization, above and beyond the effects of the sense of interest captured by attitude constructs.

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Appendix

Summary of Studies and Samples Included in the Meta-Analysis

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
Amiot et al. (2006)	Fleet staff who had been previously employed in airline companies	220	0.34	Job satisfaction (Job satisfaction)	Australia
Ashforth & Saks (1996)	Members of graduating classes of undergraduate business program	295	0.57	Job satisfaction: time 1 (Job satisfaction)	U.S.
		295	0.16	Performance: time 1 (In-role performance)	U.S.
		222	0.58	Job satisfaction: time 2 (Job satisfaction)	U.S.
		222	0.21	Performance: time 2 (In-role performance)	U.S.
		222	0.41	Lagged Job satisfaction: time 2 (Job satisfaction)	U.S.
		222	0.18	Lagged Performance: time 2 (In-role performance)	U.S.
Ashforth et al. (2007)	Business and engineering graduates from a large state university that were employed full-time following graduation	150	0.24	Performance (In-role performance)	U.S.
		150	0.48	Job satisfaction (Job satisfaction)	U.S.
Avanzi et al. (2012)	Italian law court clerks	195	0.23	Job satisfaction (Job satisfaction)	Italy
	Italian teachers	348	0.11	Job satisfaction (Job satisfaction)	Italy
Bartel (2001)	Org. members who performed community outreach for the Pillsbury Company	144	0.32	Pre-program work effort (In-role performance)	U.S.
		144	0.4	Post-program work effort (In-role performance)	U.S.
Becker (1992)	Employees in a military supply company	412	0.2	Overall prosocial behavior (Extra-role performance)	U.S.
Becker et al. (1996)	Members of the May 1993 graduating class of a large northwestern university	281	0.09	In-role performance (In-role performance)	U.S.
Bell & Menguc (2002)	A large retail insurance organization	228	0.36	OCB - Altruism (Extra-role performance)	U.S.
		228	0.38	OCB - Courtesy (Extra-role performance)	U.S.

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
Cable & DeRue (2002)	Individuals who had received their MBAs over the last 10 years at a business school in the Southeast	228	0.35	OCB - Conscientiousness (Extra-role performance)	U.S.
		228	0.49	OCB - Civic virtue (Extra-role performance)	U.S.
		228	0.29	Customers' perceptions of service quality (In-role performance)	U.S.
		136	0.13	Peer-rated citizenship behaviors (Extra-role performance)	U.S.
		187	0.44	Job satisfaction (Job satisfaction)	U.S.
		136	-0.04	Peer-rated job performance (In-role performance)	U.S.
		136	0.23	Pay raise: time 2 (In-role performance)	U.S.
Cardador et al. (2011)	Individuals from three health care occupations at a large midwestern health care organization	364	0.39	Job satisfaction (Job satisfaction)	U.S.
Carmeli et al. (2006)	Employees in four different organizations which operate in the electronics and media industries	182	0.53	Affective commitment - Joy (Affective organizational commitment)	Israel
		182	0.52	Affective commitment - Love (Affective organizational commitment)	Israel
Carmeli et al. (2007)	Employees in four different organizations which operate in the electronics and media industries	161	0.32	Job performance (In-role performance)	Israel
Carmeli et al. (2011)	Full-time employees working on advanced R&D projects in 12 high-tech organization	203	0.23	Employee knowledge sharing (In-role performance)	Israel
Chi et al. (2010)	Employees in the fast food industry	189	0.48	OCB (Extra-role performance)	Taiwan
Cho & Ryu (2009)	Korean employees in Seoul, Busan, and Gyeongnam	255	0.57	Sacrifice (Extra-role performance)	South Korea
		255	0.53	Job performance (In-role performance)	South Korea
Cho & Treadway (2011)	Bank employees	130	0.43	OCB (Extra-role performance)	South Korea
	PMBA students and employed undergraduate management students enrolled in a northeastern public university in the U.S.	135	0.48	OCB (Extra-role performance)	U.S.
Christ et al. (2003)	Teachers from all types of German schools and from eight different federal states	447	0.31	OCB towards the own qualification (Extra-role performance)	Germany
		447	0.37	OCB towards the team (Extra-role performance)	Germany

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
		447	0.43	OCB towards the organization (Extra-role performance)	Germany
Chughtai & Buckley (2010)	Full-time high school teachers in elite schools	130	0.31	In-role job performance (In-role performance)	Pakistan
		130	0.36	Error communication (In-role performance)	Pakistan
Das et al. (2008)	A mid-size, international call center	132	0.05	Performance (In-role performance)	India
DeConinck (2011a)	Sales people & sales manager using a commercial list of sales managers (500)	393	0.66	(Affective) Organizational commitment (Affective organizational commitment)	U.S.
DeConinck (2011b)	Sales people	356	0.73	(Affective) Organizational commitment (Affective organizational commitment)	U.S.
		356	0.29	Performance (In-role performance)	U.S.
De Roeck et al. (2014)	Employees of a merging hospital in Belgium	181	0.41	Job satisfaction (Job satisfaction)	Belgium
Dukerich et al. (2002)	Physicians affiliated with three health care systems	1,504	0.43	Willingness to engage in cooperative behavior: OCB (Extra-role performance)	U.S.
Edwards & Cable (2009)	Employees from four organizations (water treatment agencies)	997	0.52	Job satisfaction (Job satisfaction)	U.S.
Edwards & Peccei (2010)	A healthcare trust in the National Health Service	736	0.61	Org. involvement (Job involvement)	UK
Egold & van Dick (2014)	Occupational therapists working in 17 different establishments and clinics	246	0.31	Work engagement (Job involvement)	Germany
		246	0.32	Job satisfaction (Job satisfaction)	Germany
Evans & Davis (2014)	Alumnus of a metropolitan university situated in the southeastern U.S.	336	0.31	OCBI (Extra-role performance)	U.S.
		336	0.44	OCBO (Extra-role performance)	U.S.
Farrell & Oczkowski (2012)	Frontline employees of a major international fast food restaurant	170	0.66	Customer orientation (In-role performance)	Australia
		170	0.41	OCB (Extra-role performance)	Australia
Feather & Rauter (2004)	Public school teachers	154	0.31	Job satisfaction (Job satisfaction)	Australia
		154	0.18	OCB (Extra-role performance)	Australia
Ford & Tetrick (2011)	Employees of a small community hospital in the U.S.	171	0.2	Safety performance - Participation (Extra-role performance)	U.S.

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
		171	0.04	Safety performance - Personal protective equipment (Extra-role performance)	U.S.
Fuchs & Edwards (2012)	UK headquarters of a global research company	137	0.54	Employee prochange behavior (Extra-role performance)	UK
Fuller et al. (2006)	A health care services company comprising nurses, technicians and other support staff	205	0.24	Voice behavior (Extra-role performance)	U.S.
Gonzalez & Denisi (2009)	Employees in 30 restaurant units	291	0.58	Affective organizational commitment (Affective organizational commitment)	U.S.
Gümüş et al. (2012)	Teachers at public schools	238	0.38	Job satisfaction (Job satisfaction)	Turkey
		238	0.63	Organizational satisfaction (Job satisfaction)	Turkey
Hassan (2010)	Geographically dispersed offices of a New York State agency with 12 distinct divisions of operation	764	0.47	Job involvement (Job involvement)	U.S.
Hekman, Bigley et al. (2009)	A large not-for-profit health maintenance organization	133	0.08	Productivity (In-role performance)	U.S.
		133	0.03	Policy adherence (In-role performance)	U.S.
Hekman, Steensma et al. (2009)	A large not-for-profit health maintenance organization	193	0.10	New work behavior - Average daily secure messaging (In-role performance)	U.S.
Herrbach (2006)	Alumni who graduated from an engineering university	365	0.53	Affective commitment (Affective organizational commitment)	France
Johnson & Ashforth (2008)	Canadian subsidiary of a large international service organization in the high technology sector. Participants were regularly engaged in work duties that required direct contact with customers.	369	0.31	Customer-oriented service behavior: Self-reported (In-role performance)	Canada
		369	-0.01	Customer-oriented service behavior: Supervisor-reported (In-role performance)	Canada
		369	0.62	Customer identification (In-role performance)	Canada
Johnson et al. (2006)	Veterinarians working in a wide range of organizations	523	0.42	Job satisfaction (Job satisfaction)	U.S.
		267	0.41	Job satisfaction (Job satisfaction)	U.S.
		153	0.34	Job satisfaction (Job satisfaction)	U.S.
		103	0.37	Job satisfaction (Job satisfaction)	U.S.

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
Johnson et al. (2012)	Upper-level undergraduates enrolled in the College of Business at a large midwestern university participated voluntarily	749	0.39	OCB (Extra-role performance)	U.S.
		749	0.20	Organizational involvement (Job involvement)	U.S.
		749	0.21	Professional development (Job involvement)	U.S.
	Employees of a large midwestern university	156	0.06	Job satisfaction (Job satisfaction)	U.S.
		156	0.34	Job satisfaction (Job satisfaction)	U.S.
Jones (2010)	Employees from Green Mountain Coffee Roasters (GMCR)	162	0.51	OCB - Loyalty (Extra-role performance)	U.S.
		162	0.06	OCB - Organization (Extra-role performance)	U.S.
		162	-0.02	OCB - Coworkers (Extra-role performance)	U.S.
		162	0.05	In-role performance (In-role performance)	U.S.
		162	0.09	Customer orientation (In-role performance)	U.S.
Korschun et al. (2011)	Customer contact employees from multiple companies in the hospitality and retail industries	534	0.09	Customer orientation (In-role performance)	U.S.
Kraus et al. (2012)	Frontline employees of travel agencies	1,644	0.22	Customer orientation (In-role performance)	Germany
	Sales force in the cleaning and sanitization industry	1,813	0.21	Customer orientation (In-role performance)	U.S.
		1,813	0.13	Objective sales performance (In-role performance)	U.S.
		1,813	0.10	Objective service performance (In-role performance)	U.S.
		306	0.42	Job satisfaction: time 2 (Job satisfaction)	China
Lui et al. (2001)	Professional accountants employed in Hong Kong firms	251	0.46	Job satisfaction (Job satisfaction)	Hong Kong
Madjar et al. (2011)	12 advertising agencies in Bulgaria	157	0.29	Routine performance (In-role performance)	Bulgaria
Mael & Alderks (1993)	Soldiers (squad members, squad leaders, platoon sergeants, and platoon leaders)	1281	0.56	Job involvement motivation (Job involvement)	U.S.
		1,281	0.36	Perceived platoon effectiveness (In-role performance)	U.S.
Mael & Ashforth (1992)	Alumni of an all-male religious college	297	0.51	Satisfaction with school (Job satisfaction)	U.S.
Marique & Stinglhamber (2011)	Engineers from an engineering company in Belgium	195	0.44	Job satisfaction (Job satisfaction)	Belgium

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country	
Marique et al. (2013)	Employees of a Belgian hospital	195	0.63	Affective organizational commitment (Affective organizational commitment)	Belgium	
		159	0.29	Job satisfaction (Job satisfaction)	Belgium	
		159	0.58	Affective organizational commitment (Affective organizational commitment)	Belgium	
	Nurses and nurse's aides from another Belgian hospital	131	0.33	Job satisfaction (Job satisfaction)	Belgium	
		131	0.53	Affective organizational commitment (Affective organizational commitment)	Belgium	
		131	0.22	Affective workgroup commitment (Affective organizational commitment)	Belgium	
	Employees from an engineering company	253	0.58	Affective organizational commitment (Affective organizational commitment)	Belgium	
	Postal employees	179	0.46	Affective organizational commitment (Affective organizational commitment)	Belgium	
	Mayfield & Taber (2010)	Upper-division undergraduate students at a large northeastern public university	179	0.04	Extra-role performance (Extra-role performance)	Belgium
			179	0.02	In-role performance (In-role performance)	Belgium
226			0.13	Student-directed OCB (Extra-role performance)	U.S.	
226			0.11	Societal-directed OCB (Extra-role performance)	U.S.	
226			0.29	University-directed OCB (Extra-role performance)	U.S.	
Michel et al. (2010)	Academic staff at a German university	226	0.31	Overall satisfaction (Job satisfaction)	U.S.	
		315	0.29	Affective commitment to change (Affective organizational commitment)	Germany	
		315	0.20	Change-supporting behavior (Extra-role performance)	Germany	

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
		110	0.35	Affective commitment to change: time 1 (Affective organizational commitment)	Germany
		110	0.24	Change-supporting behavior: time 1 (Extra-role performance)	Germany
		110	0.33	Affective commitment to change: time 2 (Affective organizational commitment)	Germany
		110	0.25	Change-supporting behavior: time 2 (Extra-role performance)	Germany
Moriano et al. (2011)	Employees belonging to several Spanish public and private organizations	186	0.55	Intrapreneurial behavior (In-role performance)	Spain
Newton & Teo (2010)	Employees from five organizations	267	0.48	Job satisfaction (Job satisfaction)	(Unknown)
Norman et al. (2010)	Working adults from multiple industries and organizations	199	0.56	OCB (Extra-role performance)	U.S.
Outlay et al. (2012)	IT managers & IT professional employees in U.S.	305	0.31	Job satisfaction (Job satisfaction)	U.S.
Peters et al. (2010)	Flight attendants working for low-cost airlines and full-service airlines	154	0.65	OCB (Extra-role performance)	Thailand
Restubog et al. (2008)	Working adults employed in a variety of occupational clusters and organizations.	137	0.22	Supervisor-rated OCBs (Extra-role performance)	The Philippines
	Full-time employees of a large commercial bank	240	0.44	Supervisor-rated OCBs: time 2 (Extra-role performance)	The Philippines
Richter et al. (2006)	Work groups in five health care organizations	53	0.03	Intergroup productivity: time 1 (In-role performance)	UK
		53	0.12	Intergroup productivity: time 2 (In-role performance)	UK
Schaubroeck et al. (2013)	U.S. Army soldiers	267	0.03	Role-related performance (In-role performance)	U.S.
Schuh et al. (2012)	Employees in China	109	0.50	OCB (Extra-role performance)	China
Singh & Winkel (2012)	Graduate students of large Midwestern university	160	0.40	Helping behaviors (Extra-role performance)	U.S.
Smidts et al. (2001)	Three organization: (a) a large, nonprofit, customer services org. (b) a utilities company and (c) a bank	1,276	0.52	Job satisfaction (Job satisfaction)	U.S.
Stoner & Gallagher (2011)	Undergraduate students to help recruit full-time working individuals as respondents	123	0.23	Self-reported performance (In-role performance)	U.S.
		105	0.06	Self-reported performance (In-role performance)	U.S.

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
Tanghe et al. (2010)	Employees of 71 teams from commercially oriented service organizations	171	0.40	OCB (Extra-role performance)	(Unknown)
		71	0.48	OCB (Extra-role performance)	(Unknown)
	Students at the VU University of Amsterdam, who participated voluntarily	121	0.51	OCB (Extra-role performance)	The Netherlands
Tangirala & Ramanujam (2008)	Front-line hospital nurses about issues potentially affecting the safety of their patients	586	0.01	Voice (Extra-role performance)	U.S.
Trybou et al. (2014)	Nurse and nurse assistants working in five Belgian nursing homes	153	−0.04	In-role performance (In-role performance)	Belgium
		153	−0.05	Extra-role performance (Extra-role performance)	Belgium
Ullrich et al. (2007)	270 small travel agencies from a German franchise system	281	0.24	Customer orientation (In-role performance)	Germany
		281	0.20	Corporate citizenship behavior (Extra-role performance)	Germany
van Dick et al. (2006)	Call center agents in Germany	210	0.51	OCB (Extra-role performance)	Germany
	Teachers from all German school types	435	0.49	OCB (Extra-role performance)	Germany
	Teachers from two different German school-types (elementary school and secondary high school)	222	0.50	OCB (Extra-role performance)	Germany
	Teachers from six different German school types	520	0.36	OCB (Extra-role performance)	Germany
	Bank employees from a large regional bank in Hesse (Germany)	339	0.33	OCB (Extra-role performance)	Germany
	Employees of another German bank	107	0.30	OCB (Extra-role performance)	Germany
	Employees of a recently merged clinical hospital in Germany	387	0.30	OCB (Extra-role performance)	Germany
	A heterogeneous sample of German employees	433	0.36	OCB (Extra-role performance)	Germany
	Employees of a manufacturing company in South China.	399	0.41	OCB (Extra-role performance)	China
	Participants came from educational institutions in Germany	94	0.53	OCB: time 1 (Extra-role performance)	Germany
		94	0.41	OCB: time 2 (Extra-role performance)	Germany
		94	0.47	OCB: time 2 (Extra-role performance)	Germany
	A further education (FE) college	138	0.41	OCB (Extra-role performance)	UK
	Sales managers of travel agencies	153	0.39	OCB (Extra-role performance)	Germany
		153	−0.02	Customers' evaluations (In-role performance)	Germany
		153	0.08	Annual sales per employee (In-role performance)	Germany

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
van Dick et al. (2007)	367 school teachers and 60 head teachers	367	0.40	Follower job satisfaction (Job satisfaction)	Germany
		367	0.45	Follower OCB (Extra-role performance)	Germany
	233 school teachers and 22 head teachers	233	0.53	Follower job satisfaction (Job satisfaction)	Germany
		233	0.46	Follower OCB (Extra-role performance)	Germany
	314 travel agents in 127 travel agencies and their leaders (managers)	314	0.15	Follower organizational identification (In-role performance)	Germany
		314	0.17	Follower OCB (Extra-role performance)	Germany
		314	0.34	Follower OCB (Extra-role performance)	Germany
van Dick et al. (2008)	Bank accountants	358	0.33	OCB (Extra-role performance)	Germany
		358	0.43	Job satisfaction (Job satisfaction)	Germany
	A German travel agency organization	308	0.35	OCB (Extra-role performance)	Germany
		308	0.22	Job satisfaction (Job satisfaction)	Germany
van Dick et al. (2004a)	Bank employees from a large regional bank	358	0.43	Job satisfaction (Job satisfaction)	Germany
	Bank employees from a large regional bank	107	0.52	Job satisfaction (Job satisfaction)	Germany
	Call center agents	211	0.44	Job satisfaction (Job satisfaction)	Germany
	Employees of a clinical hospital	459	0.21	Job satisfaction (Job satisfaction)	Germany
van Dick et al. (2004b)	Teachers of all German school types	515	0.36	Job satisfaction (Job satisfaction)	Germany
van Knippenberg & Sleafos (2006)	Faculty members of a Dutch university	133	0.01	Job satisfaction (Job satisfaction)	The Netherlands
van Knippenberg & van Schie (2000)	Divisions of a local Dutch government	76	0.03	Job satisfaction (Job satisfaction)	The Netherlands
	Dutch university faculties	163	0.18	Job satisfaction (Job satisfaction)	The Netherlands
		163	0.27	Job involvement (Job involvement)	The Netherlands
Vondey (2010)	Working individuals who are employed across various industries throughout the U.S.	114	0.38	OCB - Interpersonal helping (Extra-role performance)	U.S.
		114	0.17	OCB - Individual initiative (Extra-role performance)	U.S.
		114	0.16	OCB - Personal industry (Extra-role performance)	U.S.
		114	0.61	OCB - Loyal boosterism (Extra-role performance)	U.S.
		114	0.61	OCB - Loyal boosterism (Extra-role performance)	U.S.
Vora et al. (2007)	Subsidiary managers of MNCs	209	0.47	Front-line implementer (In-role performance)	U.S.
		209	0.27	Front-line implementer (In-role performance)	U.S.

(Appendix continues)

Appendix (continued)

Article ^a	Sample	Sample size	Correlation (r)	Outcome type ^b	Country
Walker (2009)	Customer service representatives employed in a contact center	81	0.26	Performance (In-role performance)	Canada
Walumbwa et al. (2009)	Employees from a large automobile dealership	398	0.52	Voluntary learning behavior (In-role performance)	U.S.
		398	0.34	Job performance (In-role performance)	U.S.
Walumbwa et al. (2011)	A major pharmaceutical company in China	201	0.20	Job performance (In-role performance)	China
Webber (2011)	Project manager–client pairs in IT industry	53	0.12	Client satisfaction (In-role performance)	Canada
		53	0.15	Client loyalty - Word of mouth (In-role performance)	Canada
		53	0.38	Client satisfaction (In-role performance)	Canada
		53	0.20	Client loyalty - Word of mouth (In-role performance)	Canada
Wieseke et al. (2012)	Sales force in the cleaning and sanitization industry	1,548	0.29	Sales performance (In-role performance)	U.S.
		1,548	0.25	Customer satisfaction (In-role performance)	U.S.
Wolfe (2007)	Full-time employees who had a direct supervisor	208	0.53	Job satisfaction (Job satisfaction)	U.S.
		208	0.11	Overall performance - Self-direction (In-role performance)	U.S.
		208	0.04	Overall performance - Personal initiative (In-role performance)	U.S.
		208	0.23	OCB - Altruism (Extra-role performance)	U.S.
		208	0.27	OCB - Compliance (Extra-role performance)	U.S.
Zhao et al. (2014)	Employees in a large diversified company in southern China	388	0.33	OCB (Extra-role performance)	China

^a Complete references can be found in the reference section. ^b Coded variable names are denoted in parentheses.

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